Supplement: Defect Classes Surfaced

Scope

Each class below was surfaced by running the evidence engine over one author's corpus during ordinary research. Each is reported as a class that occurs and is mechanically detectable. No frequency is claimed, and no count here enters any rate the paper reports: the corpus is self-selected and its declarations were written with the tool being evaluated.

What an outside auditor can report about a stranger's repository is that two values disagree. The mechanism that produced the disagreement, and the change that closes it, are recoverable by the author and by nobody else. Each table below carries both, together with a *Missed by* row naming the checks that ran over the defect and returned success.

Class	**What is observably wrong**	**Status**
Stored value disagrees with the printed value	Nine of ten table rows disagree with the file the row is read from	Open
False mismatch under float equality	Three of thirty-nine agreements hold only under decimal comparison	Closed
Recorded hash compared against nothing	Digests recorded and never compared to the bytes on disk	Closed
Check that runs on no files	A glob matches nothing, the check exits zero, and the build is green	Open
Claims separated from the evidence they name	178 assertions marked verified, and no source in any commit	Closed locally
Reported figure that stopped reproducing	A reported count no longer reproduces from the bibliography it described	Open

Table 1: The six classes, in the order they are worked below. Status records whether a change has been made that prevents the class, not whether the instance was repaired. Two of the six are defects in a check rather than in a manuscript, which is why a report that returns success is not evidence that anything was examined.

Stored value disagrees with the printed value	
Cause	Unrecorded. Every artifact resolves at its recorded digest, so the files hold the values the run wrote. What produced the different values in the manuscript's table is recorded nowhere a check can read.
Effect	Nine of the ten rows disagree with the result file the row's value is read from. A value printed at `0.001` is stored as exactly zero, and one printed at `0.0003` is stored as `9.73e-7`. The remaining seven disagreements are not characterized here.
Detected by	Ten `metric` assertions, one per row, each addressing the same JSON Pointer in a pinned result file and compared as a decimal at the precision the manuscript printed: nine `mismatch`, one `verified`.
Missed by	Every integrity check, because the artifacts are intact. A broken pin, an absent file, and a declaration that fails to parse would each have fired, and none of the three applies. Nothing compared a file to the sentence citing it.
Mitigation	One `metric` assertion per printed value, evaluated on every build.
Cost	One declaration line per value: an artifact identifier, a pointer, and the value as the manuscript prints it.
Status: Open. The finding is recorded against the commit that showed it, and the manuscript is not yet corrected.	

Table 2: Nine of the ten rows in one manuscript's table disagree with the pinned file each row is read from. Every artifact is intact, so no integrity check fires: the disagreement is reachable only by a check that compares a file to the sentence citing it.

False mismatch under float equality	
Cause	The manuscript prints a rounded decimal and the artifact stores a binary float. Comparing the two for equality compares values that were never equal.
Effect	Three of the thirty-nine values a second manuscript prints agree only under decimal comparison at the precision printed: `0.012` against a stored `0.0122`, `0.038` against `0.0376`, `0.456` against `0.4556`.
Detected by	Thirty-nine `metric` assertions against ten pinned result files, all thirty-nine `verified`.
Missed by	Nothing, because the defect is in the checker rather than in the manuscript. Under float equality the same run over the same artifacts reports three mismatches in thirty-nine, none of which is a disagreement.
Mitigation	Reported values are carried as strings and compared with `Decimal` at the precision the manuscript printed.
Cost	Precision is taken from the manuscript, so a value printed at one significant figure is checked at one significant figure and constrains almost nothing.
Status: Closed. The comparison rule is part of the contract and a conformance fixture covers it.	

Table 3: A checker comparing a printed decimal to a stored binary float reports three findings in thirty-nine that are not disagreements. Three spurious findings in thirty-nine is the rate at which a checker is switched off, so the comparison rule decides whether the tool is usable at all.

Recorded hash compared against nothing	
Cause	Recording a digest and comparing a digest are separate operations, and only the first ran. A source carried its digest from the moment it was declared, and nothing resolved the file that digest was taken from.
Effect	Three hundred and fifty-five sources carried recorded hashes that nothing compared against the files on disk. A source edited after it was quoted therefore reported no change.
Detected by	A positive control: write a fixture, pin it, substitute one word in it, and confirm that the altered file reports a broken pin while an untouched one matches its pin. The fixture is written for the test rather than copied from the corpus.
Missed by	Quotation verification, which passed over every one of these sources. A quotation resolves against the text as it stands, so a source edited after the quotation was recorded verifies against the edit.
Mitigation	Pins are resolved before any evidence is read, and a decision computed against a broken or unpinned artifact is reported and marked non-authoritative. The positive control runs with the suite.
Cost	A digest per source, recomputed on every run, and a legitimate source update now requires the declaration to be updated with it.
Status: Closed. The control fails if the comparison is removed.	

Table 4: Digests were recorded for three hundred and fifty-five sources and compared to nothing. Quotation verification passed throughout, because a quotation resolves against the text as it stands: a digest that is never compared detects no edit at all.

Check that runs on no files	
Cause	A script collected the digests cited by every registration under one directory and verified that each still matched a file. The directory was renamed. The glob still names the old directory and matches nothing. A spell-checker exclusion elsewhere still names it too, from the same rename.
Effect	The script prints `no registrations to check` and exits zero. Over zero registrations it reports what it reports over a set that passes.
Detected by	An audit of the check itself. No assertion covers it, because the failure is in a check and not in a claim.
Missed by	Continuous integration, a pre-push hook, and a make target. All three run this script, and all three were green in this state.
Mitigation	Not implemented. The glob still names the removed directory and the script still exits zero over zero registrations. The guard is an assertion that the set to be checked is non-empty, which is the condition the policy profiles already impose on a report: every profile requires that at least one assertion was evaluated.
Cost	One assertion per check that its input is non-empty, and a decision about which checks may legitimately have nothing to do.
Status: Open. The glob matches nothing today.	

Table 5: A check whose input set is empty emits what a check whose input set passes emits. Three separate places ran this script and all three reported success, which is why a report that returns success is not evidence that anything was examined.

Claims separated from the evidence they name	
Cause	A repository was split from its working material so that a reader would not receive source extracts and caches that were not the authors' to redistribute. The split carried the claims and left the evidence.
Effect	Forty-eight committed claim files carry one hundred and seventy-eight assertions, every one marked verified. The sources those assertions name appear in no commit on any branch. A reader cloning the repository can recheck none of the one hundred and seventy-eight.
Detected by	Pin resolution, which runs before any evidence is read: forty-eight sources named, forty-eight absent.
Missed by	The stored marks. Each of the one hundred and seventy-eight assertions carries a verified flag written by a run made where the sources resolved, so the repository reports a fully verified claim set while holding no evidence at all.
Mitigation	All forty-eight sources resolve in the repository the split was made from, and all forty-eight digests match there byte for byte. A link into that tree, with an ignore entry so nothing is carried, lets verification resolve locally.
Cost	The link is local. A reader cloning the split repository resolves nothing, so the assertions are checkable by the author and not by the reader.
Status: Closed locally, open for a reader.	

Table 6: A repository split carried one hundred and seventy-eight verified assertions and left the sources behind. The stored verified marks survive the split, so the repository reports a fully verified claim set that a reader can recheck in no part.

Reported figure that stopped reproducing	
Cause	A count was reported about a bibliography, and the bibliography was then corrected, in part because the count said it needed correcting. The reported count named no version and carried no digest, so it describes a state of a path rather than a set of bytes.
Effect	An audit over a bibliography of seventy-five entries reported twenty-six whose stored metadata disagreed with the record their own identifier resolves to, across author list, author given name, title, volume and page range. The entries were repaired. The same command over the same path now reports ten. The twenty-six stands in the tool's own documentation and in the package description shipped with a release, and no longer reproduces from the file it described.
Detected by	Re-running the command that produced the count. Nothing else distinguishes a figure that has stopped reproducing from one that was never run again.
Missed by	Every check, because nothing is broken. The bibliography is correct, the audit is correct, and the command is unchanged. No pin on the bibliography existed to report the change, and no assertion bound the count to a version.
Mitigation	The reported count becomes a `metric` assertion against a pinned artifact, so it states something about a version rather than about a path.
Cost	A digest per reported figure, and a figure that has to be restated whenever the artifact it describes is deliberately changed — which, for a figure describing defects that are then repaired, is every time the work succeeds.
Status: Open. The count is not pinned.	

Table 7: An audit reported twenty-six of seventy-five entries whose stored metadata disagreed with the registry record their own identifier resolves to. The entries were repaired, and the same command over the same path now reports ten. The published count no longer reproduces, and nothing failed: the instrument worked and the artifact moved under a number that was never pinned.